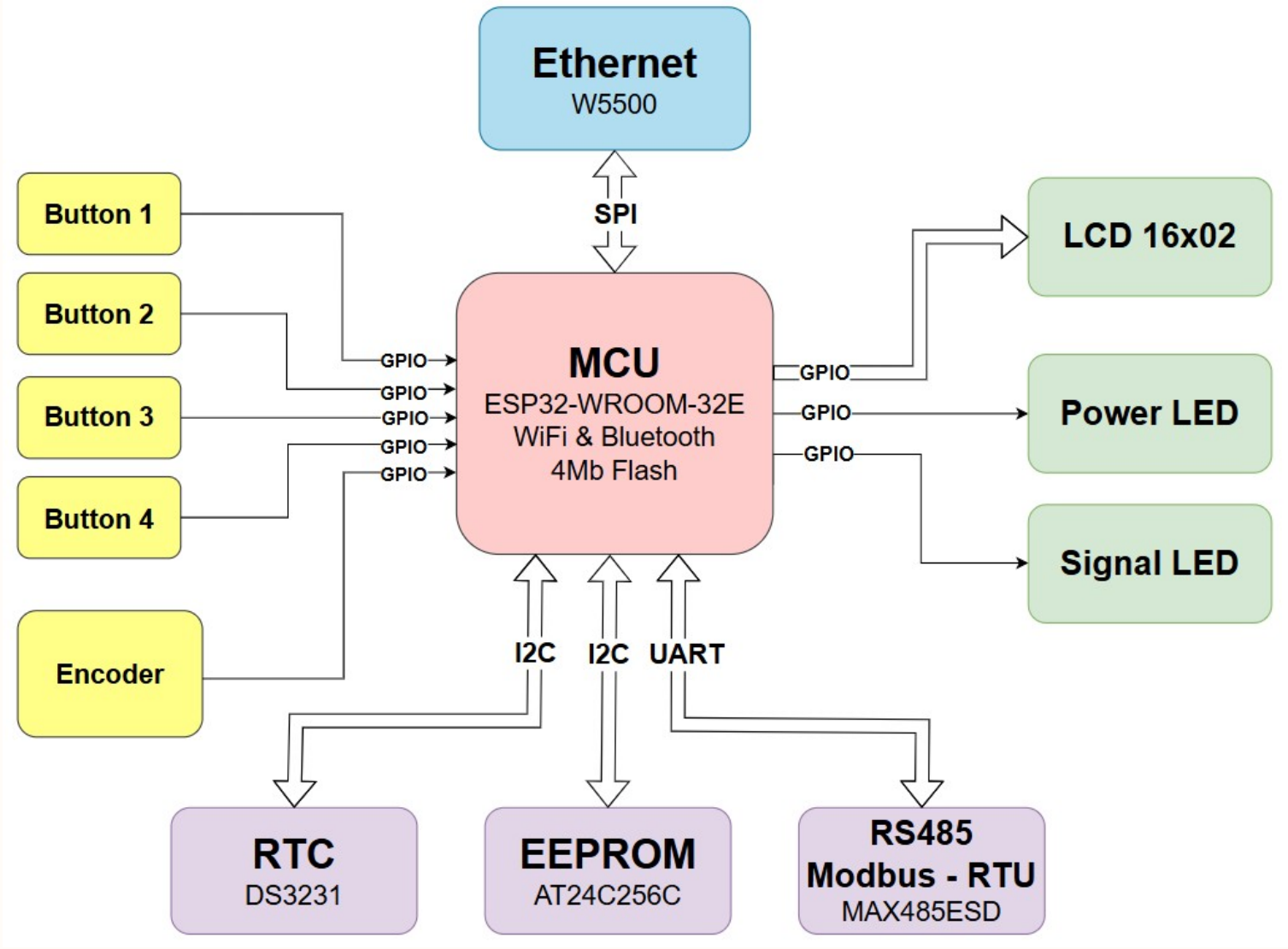
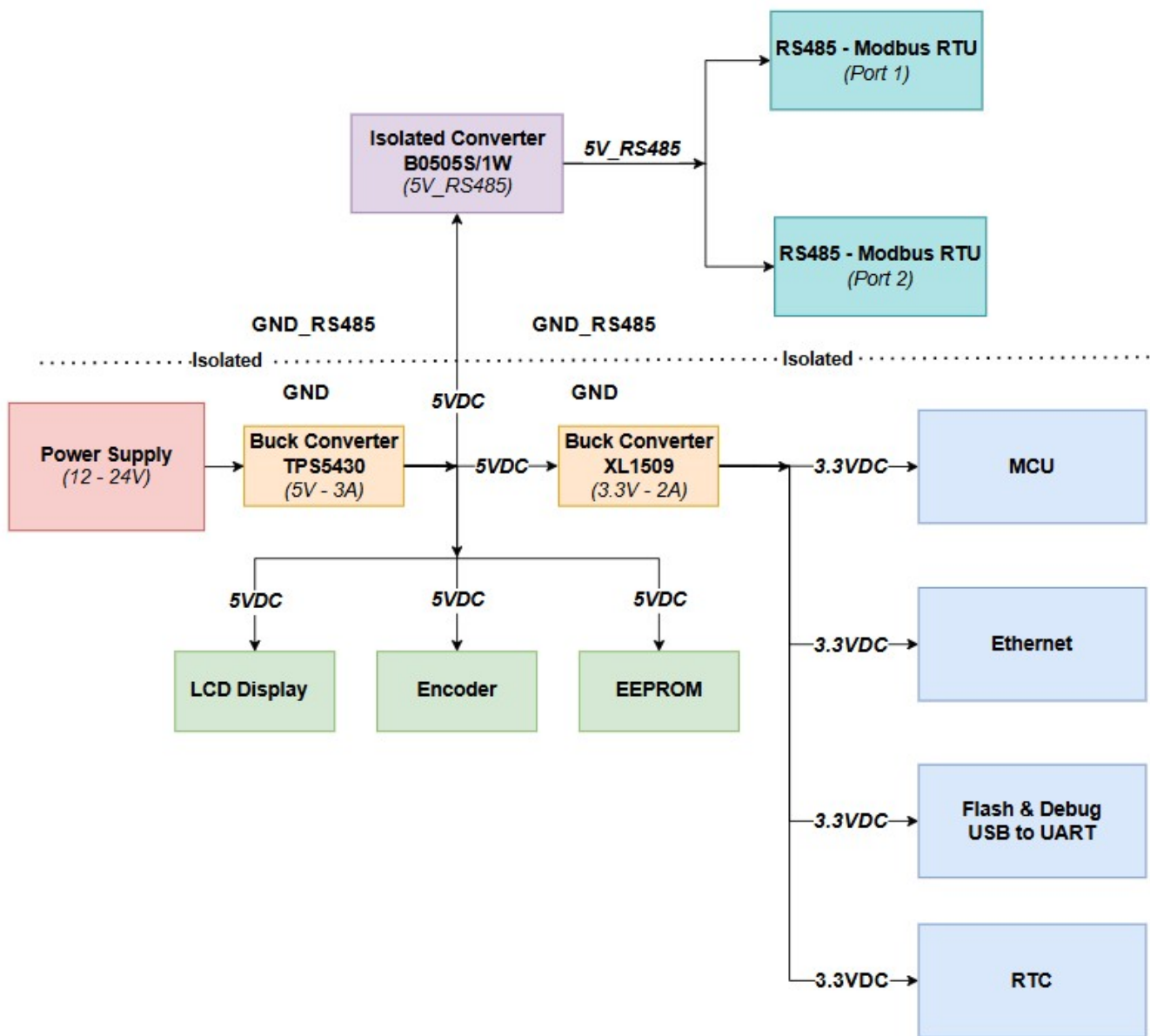


1		2		3		4	



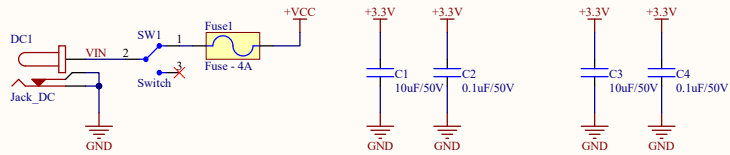
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Size	Number	Revision
A3		
Date:	2/25/2026	Sheet of
File:	D:\08 Semester 8\...\Block Diagram.Sch	Drawn By:



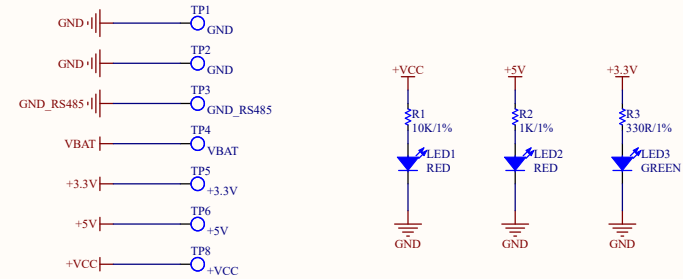
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Size A3	Number	Revision
Date: 2/25/2026	Sheet of	
File: D:\08 Semester 8\Power Tree\SchDoc	Drawn By:	

DC JACK & FILTER

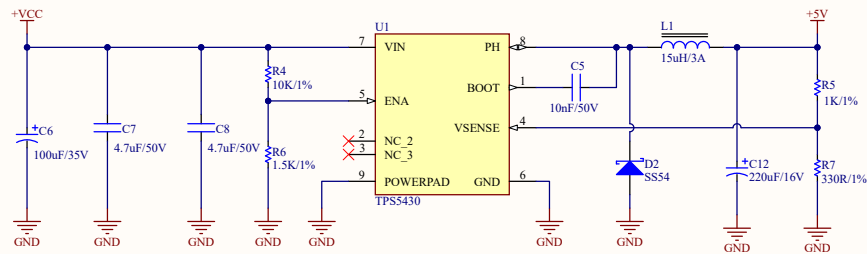
DC IN 12 - 24V



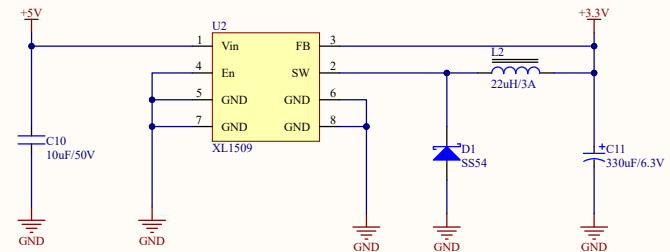
TEST POINT & POWER LEDs



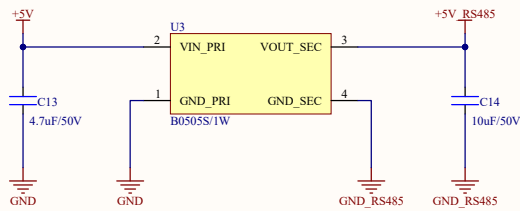
BUCK - CONVERTER - 5V



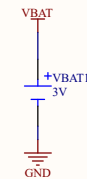
BUCK - CONVERTER - 3.3V



ISOLATED - CONVERTER 5V_RS485

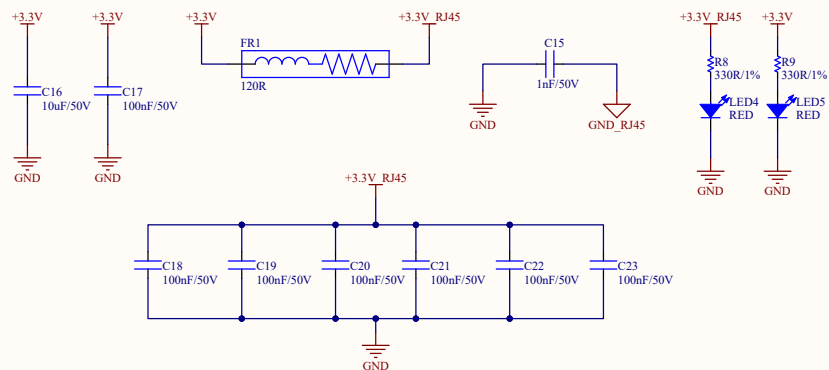


PIN 3V FOR RTC

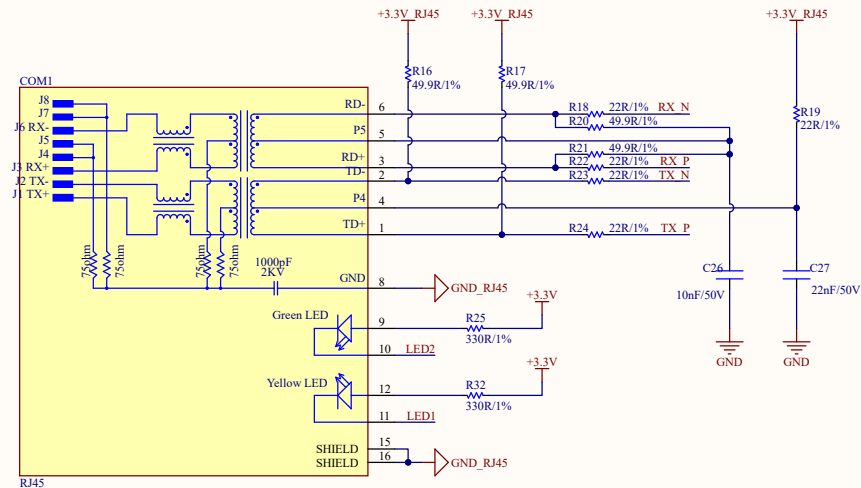


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A3		
Date:	2/25/2026	Sheet of
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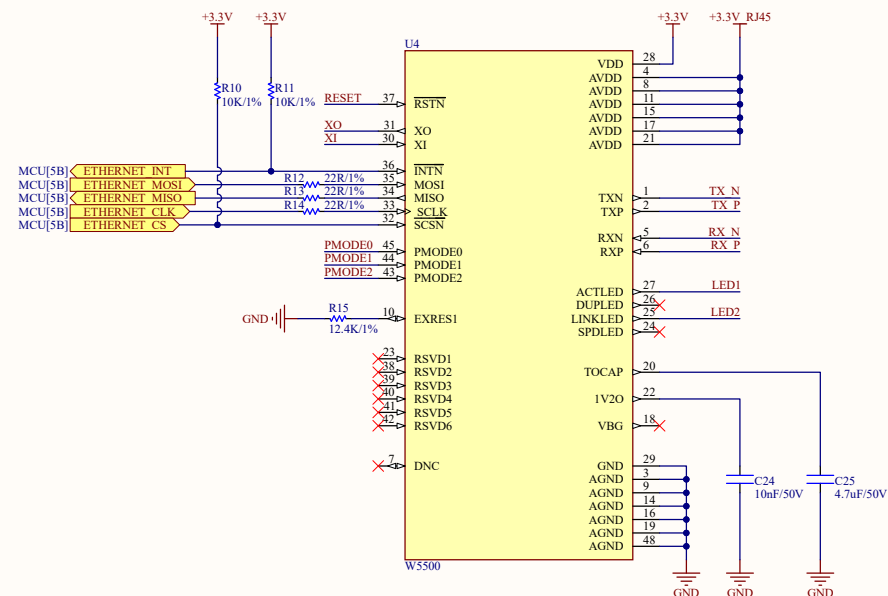
POWER FILTERs & POWER LEDs



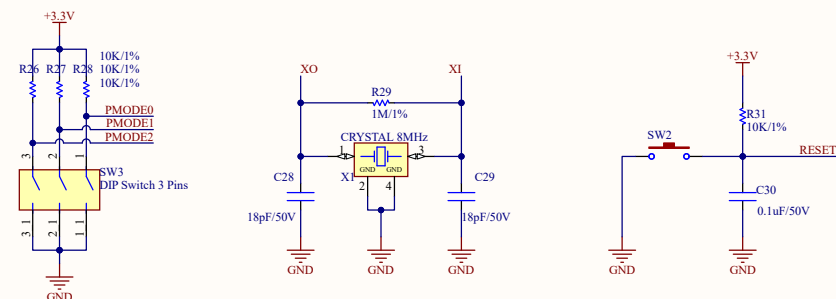
RJ45 PORT



ETHERNET - W5500



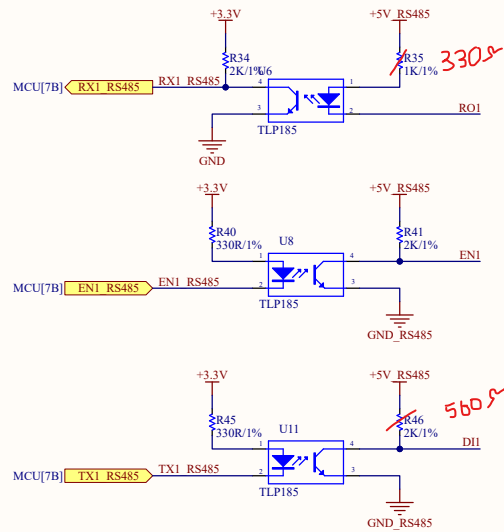
MODE SWITCH & CRYSTAL & RESET BUTTON



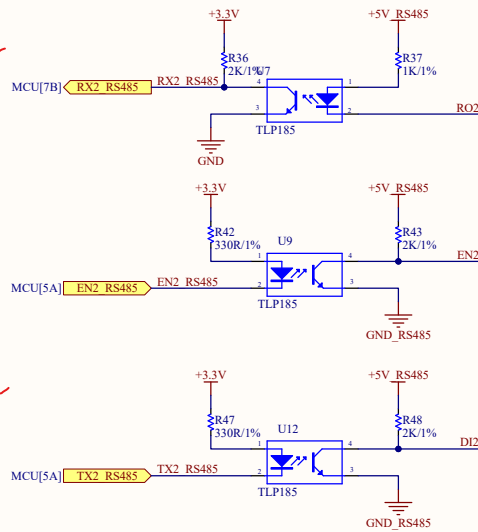
Title		
Size A3	Number	Revision
Date: File:	2/25/2026 D:\08 Semester 8\Ethernet.SchDoc	Sheet of Drawn By:

COMMUNICATE WITH MCU

PORT 1



PORT 2

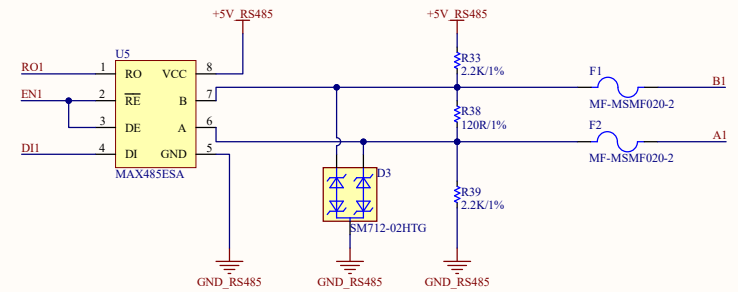


CONNECTORs & TEST POINTS

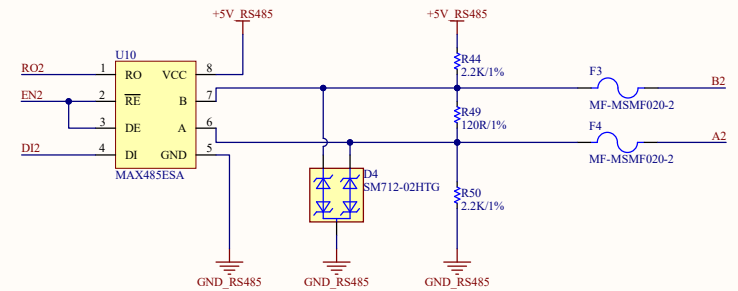


RS485 PORTS

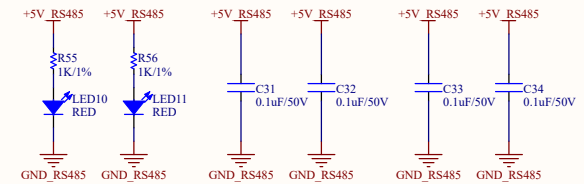
PORT 1



PORT 2

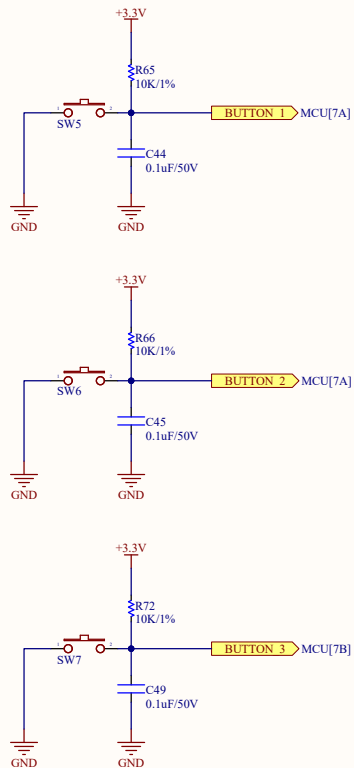


POWER LEDs & POWER FILTERS

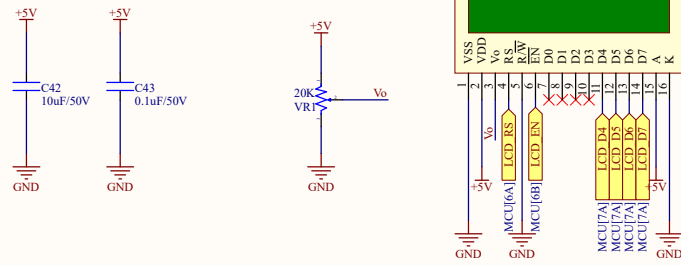


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A3		
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File:	D:\08 Semester 8\...RS485 Port.SchDoc	Drawn By:

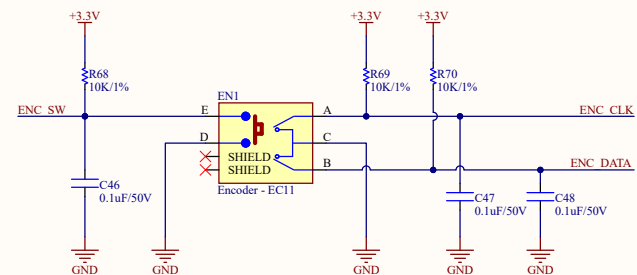
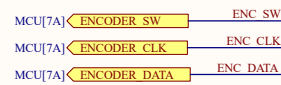
USER BUTTON



LCD 16x4A

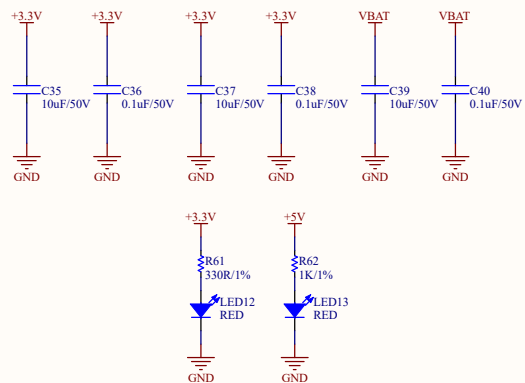


ENCODER

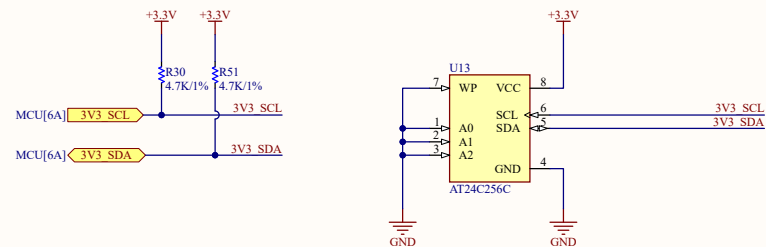


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Size A3	Number	Revision
Date: 2/25/2026	Sheet of	
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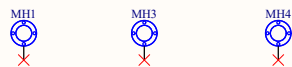
POWER FILTERs & POWER LEDs



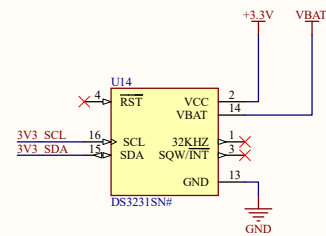
EEPROM MEMORY



MOUNTING HOLES

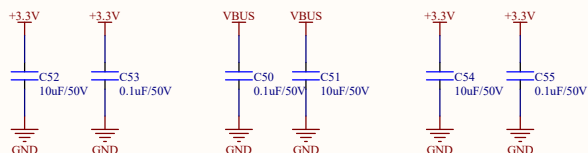


REAL TIME CLOCK (RTC)

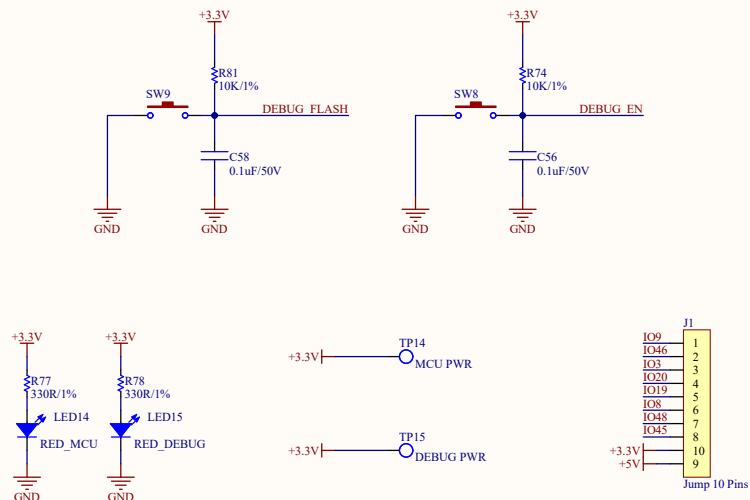


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Size	Number	Revision
A3		
Date:	2/25/2026	Sheet of
File:	D:\08 Semester 8\...RTC and EEPROM	Author By:

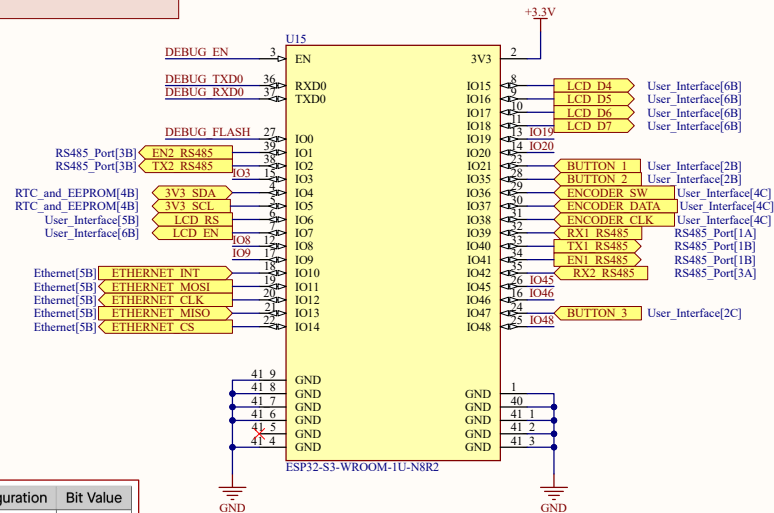
POWER FILTER & TEST POINTs



BOOT & RESET BUTTONs & POWER LEDs

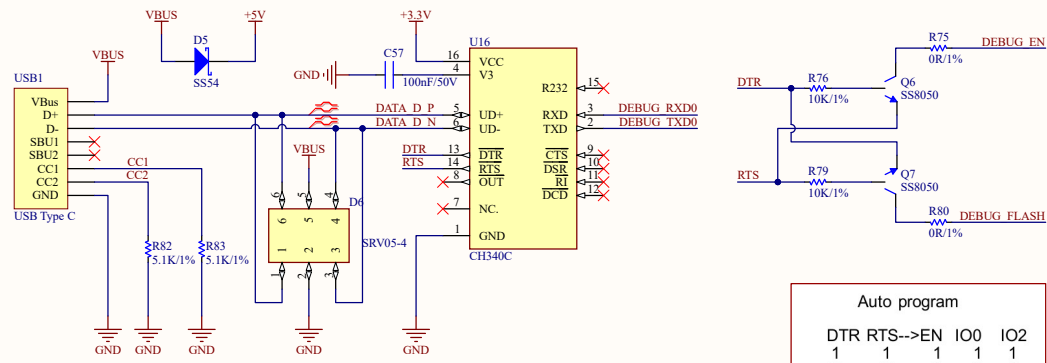


MICROCONTROLLER



Strapping Pin	Default Configuration	Bit Value
GPIO0	Weak pull-up	1
GPIO3	Floating	-
GPIO45	Weak pull-down	0
GPIO46	Weak pull-down	0

FLASH & DEBUG - USB TYPE C



Auto program

DTR	RTS	EN	IO0	IO2
1	1	1	1	1
0	0	1	1	1
1	0	0	1	1
0	1	1	0	0

Title		
Size	Number	Revision
A3		
Date:	2/25/2026	Sheet of
File:	D:\08 Semester 8\...MCU.SchDoc	Drawn By: